Department of
Computer Science and Engineering
Independent University, Bangladesh

| Sec No. | CLOs | Marks Allocated | Marks Obtained | Course Code | CSE200 |
|---|---|---|---|---|---|
| 1 | CLO1 | 40 | | **Course Title** | Intro to Programming II |
| 2 | | | | **Term** | Spring 2026 |
| 3 | | | | **Test** | Mid Term Examination |
| 4 | | | | **Time** | 90 Mins |
| 5 | | | | **Date** | February 26, 2026 |
| **Total Marks** | | 40 | | **Signature** | |
| **Instructions** | | **1.** Answer ALL questions.<br>**2.** Mobile phones or any electronic gadget is not allowed.<br>**3.** Leaving room during the exam is strictly prohibited. | | | |
| **Student Name** | | | | **Section** | |
| **Student ID** | | | | | |

## Programming Basic

**1. Study the given code snippet and write down what output they should produce.** **1.5x4= 6**

| Code Snippet | Output |
|---|---|
| a. data = list(range(16,0,-3))<br>print(data) | [16, 13, 10, 7, 4, 1] |
| b. def foo(x, y = 0, *v):<br>print(type(x), type(y), type(v), len(v))<br><br>foo('007', 3.91, 'abc', 91, 1.2) | <class 'str'> <class 'float'> <class 'tuple'> 3 |
| c. lst = [1.5, 3.9] * 3<br>for i in range(-1, -len(lst),-1):<br>print(i, lst[i]) | -1 3.9<br>-2 1.5<br>-3 3.9<br>-4 1.5<br>-5 3.9 |
| d. term = 'abracadabra_elite_u'<br><br>if 'a' in term and 'e' in term:<br>print("two vowels!")<br>if 'i' in term and 'u' in term:<br>print("two more vowels!")<br>if 'o' in term:<br>print("all vowels!") | two vowels!<br>two more vowels! |

# Output Tracing

**2. What is the output of the following program?** **3**

```
rows = 5

for i in range(1, rows + 1):
    num = 1
    for j in range(1, i + 1):
        print(num, end=" ")
        num = num * (i - j) // j
    print()
```

**Output:**

1
1 1
1 2 1
1 3 3 1
1 4 6 4 1

**3. Write down the order of line numbers (comma separated) in which the following code is executed. Also write the output of the program.** **2 + 1 = 3**

```
1  from sympy import isprime
2  def oddev(data):
3    if data % 2 == 0:
4      return 'even'
5    else:
6      return 'odd'
7  def prime(data):
8    if isprime(data):
9      return 'prime'
10   else:
11     return 'not prime'
12 nums = [7, 12]
13 for i in nums:
14   print(oddev(i), prime(i))
```

**Order of line numbers:**
1,12,13,14,2,3,5,6,14,7,8,9,14,
13,14,2,3,4,14,7,8,10,11,14,13

**Output:**

odd prime
even not prime

**4. What is the output of the following program?** **2**

```
lst =['apple', 'orange', 'mango', 'papaya', 'banana']

for i in range(len(lst)):
  count = 0
  for j in range(len(lst[i])):
    if lst[i][j] in 'aeiou':
      count += 1
  print(f'{lst[i]} has {count} vowels')
  i +=1
```

**Output:**

```
apple has 2 vowels
orange has 3 vowels
mango has 2 vowels
papaya has 3 vowels
banana has 3 vowels
```

---

## Code Refactoring

**5.** Rewrite the program in **Question 4** by implementing the **for** loop with **while** loop. The program must produce the same output. **5**

```
lst =['apple', 'orange', 'mango', 'papaya', 'banana']
i = 0
while i < len(lst):
  count = 0
  j = 0
  while j < len(lst[i]):
    if lst[i][j] in 'aeiou':
      count += 1
    j += 1
  print(f'{lst[i]} has {count} vowels')
  i += 1
```

---

## Error Tracing

**6. Follwoing program has syntax and logical errors. Correct each line in which you find such errors and write the corrected code in the corresponding empty space (One error is corrected for your reference).** **4**

| Program with Error(s) | Corrected Program |
|---|---|
| `info = {'P', 'y', 't', 'h', 'o', 'n', 'I', "s", 'M', 'y', 'l', 'i', 'f', 'e', '!'}` | `info = ['P', 'y', 't', 'h', 'o', 'n', 'I', "s", 'M', 'y', 'l', 'i', 'f', 'e', '!']` |
| `i == 0` | `i = 0` |
| `count = '0'` | `count = 0` |
| `while i < len(info):` | `while i < len(info):` |
| `  if info[i] == 'aeiouAEIOU':` | `  if info[i] in 'aeiouAEIOU':` |
| `    del.info[i]` | `    del info[i]` |
| `    count + = 1` | `    count += 1` |

| | |
|---|---|
| `i += 1` | `i += 1` |
| `print(f'After deleting [count] vowels the list is: [info]')` | `print(f'After deleting {count} vowels the list is: {info}')` |

## Program Writing

**7. Given a square matrix, compute the *summation and average* of the diagonal of the matrix. The row-column condition for finding a Diagonal is, row = column.**
**USE While loop.** **3.5**

Example input/output:

For the given matrix mat, the output will be: sum = 8 average = 2.0

mat = [[1,2,3,4], [4,3,2,1], [1,2,3,4], [4,3,2,1]]

**Write your code here:**

```
row = 0
sum = 0
while row < len(mat):
  sum += mat[row][row]
  row += 1
print(f'sum = {sum}, average = {sum / len(mat)}')
```

**8. Write a simple program to sort the following array of numbers in descending order.** **1.5**

Data = [12, 4, 25, 12, 1, 9, 20.5, 99.12, 6, 13, 12]

**Your code:**

```
data.sort()
data.reverse()
print(data)
```

**9. Complete the following program as instructed.** **12**

```
# the following list holds marks for four students as follows:
```

```
# first two marks are quiz1 and quiz2 marks. Both are out of 10.
# thrid mark is the mid exam makrs. Out of 30.
# forth mark is the final exam marks. out of 50.
# So the total marks for the exam is 100.
marks = [[9.5, 6, 22, 31.5],
         [6.5, 8.1, 25, 39.5],
         [8.5, 9, 27.5, 29],
         [10, 8.5, 11, 48.4]]

# Considering the above list of marks, complete / write python code to
accomplish the following tasks:

#TASK 1: complete the follwoing function that will take the marks list
as argument and
# print the highest marks achieved by any student in Quiz1, Quiz2, Mid
and Final exam
# In case of given marks, Q1 has highest 10, Q2 is 9, Mid is 27.5 and
final is 48.5 among all the students.
# the function should return all the highest marks in sequence
# [Q1, Q2, mid, final]
def findAndReturnHighest(marks):
```

```
#TASK 2: Complete the follwoing function that will take the marks list as argument and
# calcuate the total marks obtained by each student.
# It will print the total marks list and return the list of total marks.
def totalMarksOfEachStudent(marks):
```

```
#TASK 3: Complete the follwoing function that will take the list of total marks as argument
# and calculate and print the corresponding grade. The grade should be calcuated
# according to IUB standard grading system. For each student it will print the following:
# Achieved mark: ?? - Grade: ??
def calcualteAndPrintGrade(stTotalMarks):
```

```
#Driver code:
#Calling the above functions in sequence

findAndReturnHighest(marks)
totalmarks = totalMarksOfEachStudent(marks)
calcualteAndPrintGrade(totalmarks)
```

***Sample output:***

```
The above function call should produce the following results
10 9 27.5 48.4
[69.0, 79.1, 74.0, 77.9]
Achieved mark: 69.0 - Grade: C
Achieved mark: 79.1 - Grade: B-
Achieved mark: 74.0 - Grade: C+
Achieved mark: 77.9 - Grade: B-
```

**SOLUTION:**

```
marks = [[9.5, 6, 22, 31.5],
         [6.5, 8.1, 25, 39.5],
         [8.5, 9, 27.5, 29],
         [10, 8.5, 11, 48.4]]

def findAndReturnHighest(marks):
  q1 = list()
  q2 = list()
  mid = list()
  final = list()
```

```
  for record in marks:
    q1.append(record[0])
    q2.append(record[1])
    mid.append(record[2])
    final.append(record[3])

  q1.sort()
  q2.sort()
  mid.sort()
  final.sort()

  # print the highest of each exam
  print(q1[len(q1)-1], q2[len(q2)-1], mid[len(mid)-1], final[len(final)-1])

  # return all the highest of each exam
  return [q1[len(q1)-1], q2[len(q2)-1], mid[len(mid)-1], final[len(final)-1]]


def totalMarksOfEachStudent(marks):
  stmarks = list()

  for record in marks:
    sum = 0
    for mark in record:
      sum += mark
    stmarks.append(sum)

  print(stmarks)
  return stmarks

def calcualteAndPrintGrade(stTotalMarks):
  for record in stTotalMarks:
    if record >=91 and record <= 100:
      print(f'Achieved mark: {record} - Grade: A')
    elif record >=85 and record <= 90:
      print(f'Achieved mark: {record} - Grade: B+')
    elif record >=81 and record < 85:
      print(f'Achieved mark: {record} - Grade: B')
    elif record >=75 and record <= 80:
      print(f'Achieved mark: {record} - Grade: B-')
    elif record >=70 and record < 75:
      print(f'Achieved mark: {record} - Grade: C+')
    elif record >=65 and record < 70:
      print(f'Achieved mark: {record} - Grade: C')
    else:
      print(f'Achieved mark: {record} - Grade: F')

#Driver code:
#Calling the above functions in sequence

findAndReturnHighest(marks)
totalmarks = totalMarksOfEachStudent(marks)
calcualteAndPrintGrade(totalmarks)
```